



Your Backups Will Be Attacked. Will They Survive?

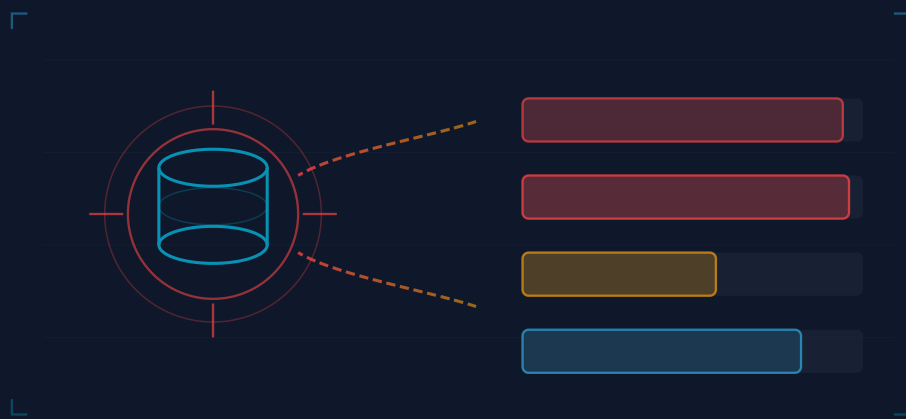
**Filesystem-Native Immutable Backup That Attackers
Cannot Encrypt, Delete, or Compromise**

Verified recovery in minutes, not hours, powered by btrfs architecture

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Paying Doesn't Buy Recovery

84% of Ransom Payers Did Not Fully Recover Their Data



84% of organizations that paid the ransom did not fully recover their data (Sophos 2024). Only **7% of payers** retrieved everything (Hiscox). **80% of payers were attacked again** within the following year (Halcyon 2024).

Payment is not an exit. It is a partial-credit ticket to a system that may still be encrypted, may still be backdoored, and is statistically more likely to be hit again. The reports your security team already reads spend most of their pages on prevention and detection; the line that matters for backup strategy is the one above.

That is why **64% of organizations now refuse to pay** (Verizon DBIR 2025), up from 50% in 2022. Refusal only works if backups survive. Everything in the rest of this deck is a function of that single dependency.

The Backup Targeting Epidemic

Attackers Destroy Backups in 94% of Incidents and 57% Succeed

94% of organizations hit by ransomware had attackers attempt to compromise their backups (Sophos 2024), and **57% of those attempts succeeded**. When backups are compromised, recovery costs jump from \$375K to **\$3M**, an **8x increase**. Recovery time extends from under one week to over a month, and **67% of victims pay the ransom** versus 36% with intact backups.

Veeam's 2025 research corroborates: **89% of ransomware attacks target backup repositories** and **73% of attempts succeed**. And **88% of ransomware breaches hit SMBs** (Verizon DBIR 2025), the exact segment that lacks specialized security headcount to harden backup infrastructure properly.

This pattern reveals a shift in attacker methodology. Backup destruction is now the primary objective of the initial intrusion phase, not an afterthought. Operators harvest backup administrator credentials, map backup infrastructure during reconnaissance, and time encryption operations to coincide with backup rotation windows. Attack sophistication has scaled in direct proportion to basic backup adoption: as more organizations maintain backups, attackers adapted by targeting the backups themselves.

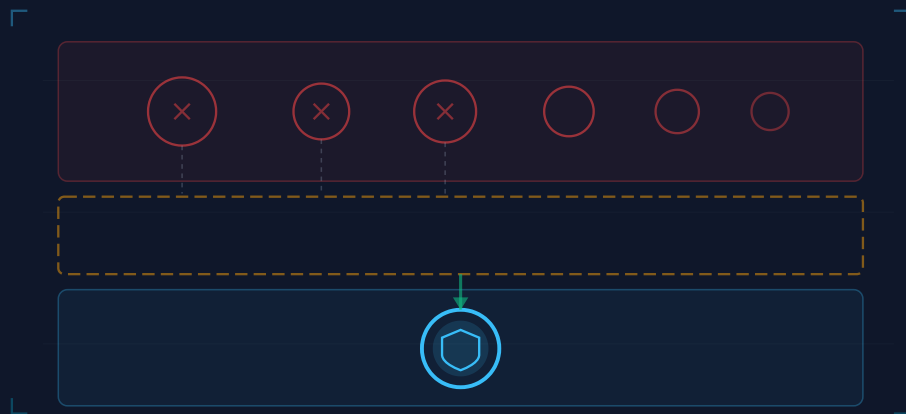
When backups are compromised, recovery costs jump 8x, from \$375K to \$3M. Backup integrity is the single largest determinant of ransomware incident cost.

The economic model is unambiguous. Intact immutable backups: \$375K average recovery cost, recovery time under one week, 36% ransom payment rate. Compromised backups: \$3M, recovery beyond a month, 67% ransom payment rate.

The difference is not endpoint protection or incident response quality. It is whether the backup system survived the attack. Against an average recovery cost of \$1.53M to \$2.73M, any annual immutable-backup investment under \$100K is self-evidently ROI-positive.

The Immutability Gap: Bolt-On vs. Native

Every Tier 1 Vendor Uses Bolt-On Immutability and Inherits a Structural Vulnerability



All Tier 1 vendors use **bolt-on immutability**: immutable flags, object locks, or proprietary appliances layered on top of traditional backup architectures. None offers filesystem-native immutability on commodity hardware. The structural differentiation gap cannot be closed without re-architecting the platform.

Veeam's Hardened Linux Repository uses XFS with the Linux `chattr +i` immutable flag, an OS-level attribute that any process with root privilege can remove. Veeam mitigates with single-use credentials and time-manipulation monitoring, but the underlying weakness stands: **root compromise defeats all protections**. Veeam itself recommends physical servers with camera surveillance, an implicit admission that software alone is insufficient.

The approach requires manual OS hardening, SSH lockdown, physical server deployment, and ongoing security patching, operational burden that most SMB and mid-market IT teams lack the headcount to maintain.

Rubrik's Atlas is a proprietary appliance-locked filesystem; its \$10M Ransomware Recovery Warranty requires a 250TB minimum and Enterprise Edition licensing, excluding most organizations below that threshold. S3 Object Lock provides strong immutability in Compliance mode, but recovery requires pulling data across the network, unsuitable for rapid restore. Governance mode can be bypassed by anyone holding the right IAM permissions.

Air-gapped vaults offer the strongest isolation and the worst recovery speed: **hours to days** for tape retrieval, with backup frequency capped at intervals that create RPO gaps of 24 hours or more.

All competing immutability approaches share the same structural weakness: protection is added on top of traditional storage. Filesystem-native immutability makes protection a property of the storage itself, with no trust boundary that stolen credentials can bypass.

The Immutability Gap (continued)

Tier 2 Adjacencies and the Filesystem-Native Alternative

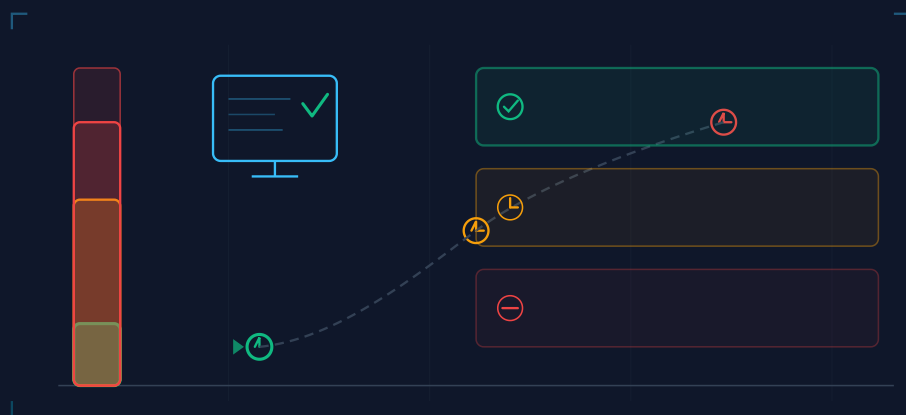
Tier 2 security-adjacent players are complementary, not competitive. CrowdStrike has no native backup capability and partners with Rubrik and Commvault. SentinelOne's ransomware rollback uses Windows VSS snapshots scoped to endpoints with **4-hour RPO**.

Neither addresses the backup repository attack surface.

The architectural alternative is filesystem-native: copy-on-write storage where snapshots are read-only by construction and deletion requires the orchestrator's authority over the storage layer, not root on the host. CRC-32C checksumming of both data and metadata gives **btrfs scrub** a way to detect silent corruption that traditional backup systems miss entirely, directly addressing the **31% backup failure rate** At-Bay reports during actual ransomware recovery attempts. The trust boundary moves from "any root process" to "the storage layer itself," which is the only boundary stolen credentials cannot cross.

Recovery Speed: Minutes, Not Days

Boot-and-Verify Architecture vs. Traditional Restore Workflows



Average ransomware-related downtime is **24 days** at an estimated **\$88,000/hour** in downtime cost (Veeam). Even "instant recovery" solutions from Tier 1 vendors do not deliver what the name implies. Veeam's Instant Recovery runs VMs from backup storage at degraded performance; full 5TB VM restores take **4 to 6 hours**.

At \$88,000/hour, every recovery hour adds six figures to incident cost. Air-gapped vaults, considered the gold standard for isolation, deliver the worst recovery: hours to days for tape retrieval, with limited backup frequency creating 24-hour-plus RPO gaps.

Mounting a btrfs snapshot as a live filesystem is a metadata operation, not a data-copy operation. There is no rehydration, no extraction, no reconstruction. The system pivots to the snapshot's point-in-time state through a pointer update. Combined with btrfs reflinks, which create instant space-efficient copies of entire subvolumes, restore time becomes independent of dataset size. No competitor can match that workflow without changing their underlying storage architecture.

System portability extends the advantage. Btrfs send/receive replicates any snapshot to any other btrfs filesystem regardless of underlying hardware. Recovery can be tested on different hardware or OS configurations before committing to a production restore.

Cloning a snapshot into an isolated test environment, booting it, verifying applications run, and confirming data integrity eliminates the scenario where organizations discover corruption only during an actual crisis. That scenario occurs in **31% of recovery attempts** (At-Bay) and **60% of data backups are incomplete** per industry surveys. Boot-and-verify converts backup integrity from an assumption into a documented fact.

Compliance and Insurance: From Best Practice to Legal Requirement

SEC, DORA, NIS2, and the 73 to 75% Insurer Mandate



Documented, tested backup and recovery is now a legal requirement across multiple jurisdictions, not a best practice that organizations can defer.

The **SEC's cybersecurity disclosure rule** (effective December 2023) requires public companies to report material cybersecurity incidents within **4 business days** on Form 8-K, with fines up to \$25M for non-compliance. Organizations that cannot recover quickly enough to assess materiality within the disclosure window face the ransomware damage and the regulatory penalty.

The **EU's DORA regulation** (enforceable January 2025) mandates ICT risk management frameworks for all financial entities, with fines up to **2% of global turnover**. **NIS2** covers 18 critical sectors with penalties up to \$10M or 2% of turnover. The EU Cyber Resilience Act (November 2024) adds requirements for digital products and services.

Cyber insurers moved in lockstep. **73 to 75% of carriers now mandate immutable, air-gapped backup systems** as non-negotiable coverage prerequisites (AI Readiness Lab 2025).

Insurer questionnaires now ask directly: "Do you have immutable, tested backups with documented restoration capabilities?" Claims have been denied when policyholders attested to controls they had not actually implemented. The cyber insurance market reached \$15B to \$16.6B in 2024 and is projected to \$32B by 2030 (Munich Re); every dollar of premium growth increases pressure to prove backup immutability.

73 to 75% of cyber insurers now mandate immutable backups. Organizations with proven immutable backup systems report **20 to 50% premium reductions**; organizations without them face coverage denial.

Btrfs's built-in integrity verification maps directly to these requirements. Every snapshot is checksummed, timestamped, and verifiable. Compliance reports proving immutability enforcement, successful recovery tests, and retention adherence can be generated as a

Object First, Pure SafeMode, ExaGrid: The Purpose-Built Approach

Hardware Appliances Solve the Storage Layer and Leave Recovery Unaddressed

A newer class of competitors targets the immutability gap directly: purpose-built backup storage appliances designed to deliver out-of-the-box immutability without the complexity of hardening general-purpose infrastructure.

Object First was founded by Veeam co-founders. Their **Ootbi** (Out-of-the-Box Immutability) is a purpose-built hardware appliance for Veeam environments, using S3-compatible object storage with hardware-enforced immutability. The appliance is "set and forget": no Linux hardening, no security patching of the storage OS.

It is **Veeam-exclusive**, meaning it only works with Veeam Backup & Replication. Recovery still relies on Veeam's standard restore workflow: data must be read from the appliance and written back to production storage, a process measured in **hours for large environments**.

Pure Storage SafeMode delivers hardware-enforced immutability through FlashArray, with Pure1 management. The approach is technically strong; the constraint is array lock-in, not architecture.

ExaGrid uses a tiered model where immutability applies only to the retention tier, not the primary landing zone, so recently written backups remain mutable for a window that varies by configuration. Standard restore workflow applies.

The shared limitation across purpose-built competitors: they solve the storage layer of the immutability problem and leave recovery speed, boot-and-verify testing, and filesystem-level integrity verification unaddressed. Filesystem-native immutability collapses all three into one architecture on commodity hardware, with no appliance lock-in.

Hyperscaler and Cloud-Native Limitations

AWS Backup Vault Lock, Azure Immutable Blob, and the Single-Cloud Trap

The three major cloud providers each offer native immutable backup. The services integrate cleanly within their respective ecosystems and inherit a structural limitation that creates the multi-cloud, on-premises-capable opening.

AWS Backup Vault Lock enforces WORM retention on AWS Backup vaults. In compliance mode, even the AWS account root user cannot delete data during the retention period. Strong protection for AWS-native workloads; recovery requires reading data across AWS's network, creating latency-dependent recovery times that scale with data volume.

Backup Vault Lock protects only AWS-hosted data. Organizations running hybrid or multi-cloud environments, **73% of enterprises** per Flexera's 2024 State of the Cloud report, cannot use Vault Lock for on-premises servers, Azure workloads, or other providers' resources.

Azure Immutable Blob Storage offers time-based retention and legal hold policies for Azure Blob containers. Immutability is strong within Azure and limited to Azure-hosted data.

Recovery requires transfer from Blob Storage to production compute, adding network-dependent latency. Azure's version-level immutability requires a flat namespace structure, which can conflict with existing data organization and force reorganization before implementation.

The fundamental limitation shared by all hyperscaler immutable backup services is ecosystem lock-in. AWS protects AWS. Azure protects Azure. Google protects Google.

No cloud provider offers immutable backup that spans on-premises, multi-cloud, and edge environments from a single control plane. For the 73% of enterprises operating in hybrid or multi-cloud configurations, cloud-native immutability covers only a fraction of the total backup surface. Filesystem-native immutability runs anywhere Linux runs: on-premises bare metal, any cloud provider's compute, or edge sites, with consistent guarantees regardless of where the data lives.

ROI: Prevention Economics vs. Recovery Economics

The 8x Cost Gap, the 24-Day Downtime Tail, and the Insurance Discount



The ROI argument for immutable backup does not require optimistic assumptions. It survives pessimistic ones.

The single-incident delta between compromised and intact backups is **\$2.625M** (\$3M vs. \$375K, Sophos 2024). That number is the direct recovery cost only.

Add the downtime tail: a 24-day average incident at \$88,000/hour adds roughly **\$50M** in lost-business cost on top of the recovery bill. A recovery-speed improvement of even a single order of magnitude eliminates that tail.

Add the insurance layer: organizations with proven immutable backup report **20 to 50% premium reductions** (AI Readiness Lab 2025). Premium savings on a mid-market cyber policy frequently exceed the annual cost of the backup infrastructure that earned them.

Scenario	Recovery Cost	Recovery Time	Ransom Payment Rate	Insurance Impact
Compromised backups	\$3M average	Over 1 month	67% pay ransom	Premium hike or denial
Intact immutable backups	\$375K average	Under 1 week	36% pay ransom	20 to 50% premium cut
Delta	\$2.625M saved	Weeks saved	31 points lower	Material savings

The calculation is insensitive to attack-probability assumptions. Even at a modest 10 to 15% annual probability of a ransomware attack, expected value of immutable backup is positive.

With **64% of organizations now refusing to pay ransom** (Verizon DBIR 2025), intact backup is increasingly the only recovery mechanism on the table. The question is not whether immutable backup pays for itself. It is whether the organization can absorb the \$2.625M gap of not having it.

Your Backups Will Be Attacked. Make Sure They Survive.

Filesystem-native immutable backup that attackers cannot encrypt, delete, or compromise. Verified recovery in minutes, not hours.

94% of ransomware attacks target backups. 57% succeed.
The difference between a \$375K incident and a \$3M catastrophe is backup immutability.

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Immutability included. Not add-on.